Onboarding reviewers should judge the exceptions, not re-type Arabic documents

Qareena reads a merchant's national ID, business license and tax card from phone photos. It checks every field against written rules and against the other documents, then hands the reviewer only what needs a human, with the photo region as evidence.

THE USER

Primary: the ZainCash merchant-onboarding reviewer who approves or returns a merchant's file.
Secondary: the merchant, who photographs the documents on a phone and needs to know straight away when a photo can't be read.

MVP packet (per the brief): national ID, business license (Chamber of Commerce ID or practice certificate) and tax card. ZainCash's published merchant list also asks for a residence card and a lease. Document types are data (schema + rulebook), not code.

THE PROBLEM TODAY

1. **Reading by hand.** Arabic fields, some of them handwritten, are re-read and re-typed from blurry, tilted, glare-covered photos.
2. **Comparing by eye.** Names and numbers are matched across three documents. Spelling variants (عبدالله / عبد الله) look like conflicts, and real conflicts (حسين / حسن) look like typos.
3. **Bad photos found late.** The file goes back to the merchant and onboarding stalls. In MENA, 42% of ID-check rejections are low-quality images (Shufti, 2.1M checks, Jan–Apr 2025).

WHAT QAREENA RETURNS: ONE OF THREE OUTCOMES, ALWAYS WITH EVIDENCE

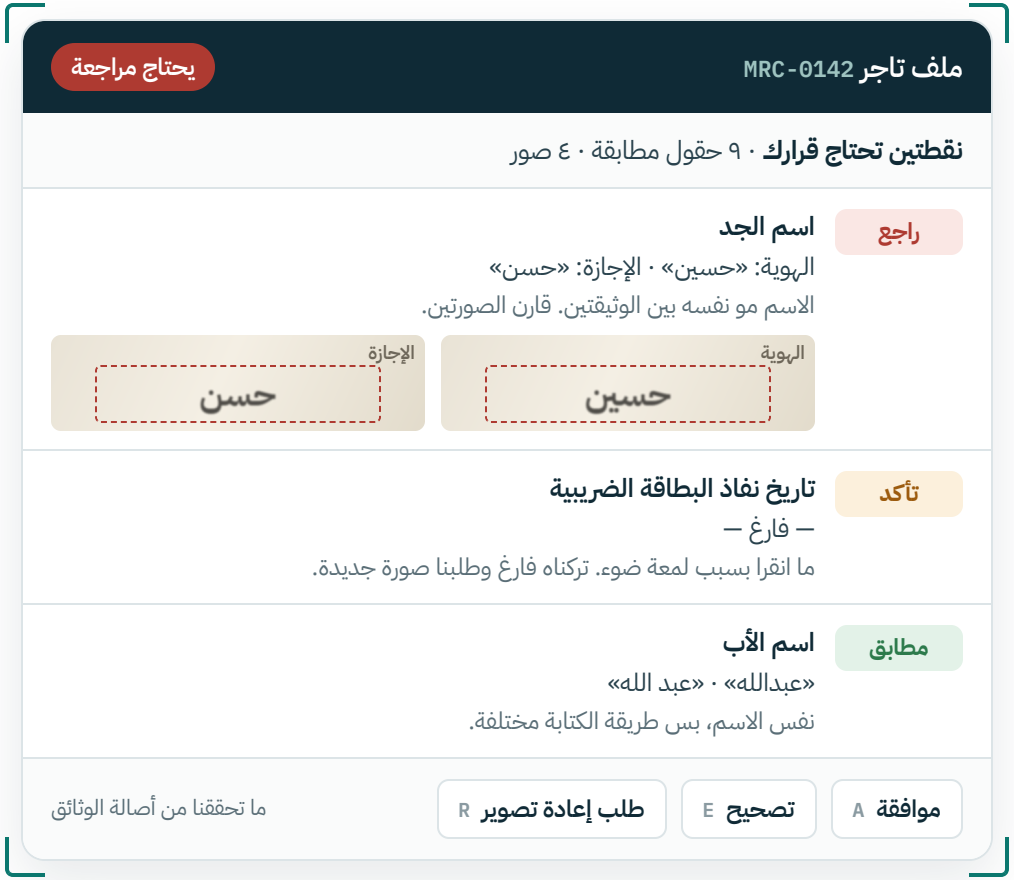
✓ Checks passed
All required fields read with evidence; rules and cross-checks clean.

! Human review
Only the fields that need a decision, each with crop and reason.

↻ Retake
Specific guidance to the merchant in Iraqi Arabic, at capture.

Scope. Qareena reads, validates and cross-checks. Face matching, liveness and forgery detection are out of scope (per the brief), so "checks passed" never means "identity verified".

Builder. Momen Saeb, who designed and built a local OCR archiving system for Iraqi court documents.

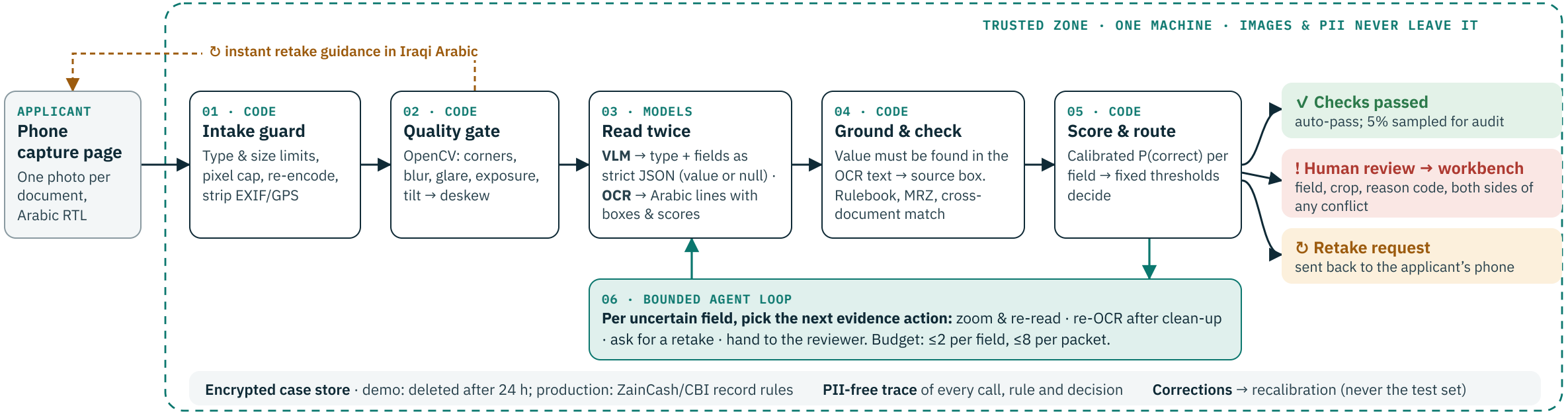


Target reviewer view (illustrative, fictional data). Every flag links to its photo region. Confidence is shown as action bands, not percentages.

User evidence. Week 1: interview an onboarding reviewer and ask ZainCash for a session with their team. Baseline minutes per file will be measured, not assumed.

A deterministic pipeline, one bounded agent loop, and a human for every exception

Models read, and code makes every decision that must be repeatable. As “Building effective agents” advises, we loop only where the number of steps is unknown: gathering more evidence for an uncertain field, with a budget and a stop rule.



Code decides (deterministic, unit-tested)

- File checks, quality thresholds, retention
- Rulebook: required fields, formats, expiry, age; ICAO 9303 MRZ check digits
- Cross-document: numbers exact; names **same / variant / missing part / conflict**
- Thresholds and routing; only grounded fields are accepted

The models and the agent loop do

- **Read:** classify each photo, extract fields, null when unreadable
- **Re-read:** a zoomed crop when readers disagree. Reads must agree to count
- **Diagnose:** glare, blur, cut corner or finger → the right retake message
- **Explain:** reviewer summary assembled from reason codes, not free text

Never allowed

- Fill a field it could not read: blank and flagged beats invented
- Accept on its own say-so, or override a rule or conflict
- Obey text inside a document: it is data, and the model has no tools
- Claim identity, authenticity or forgery results

Models chosen by measurement; fictional documents photographed badly on purpose

Everything in the live path runs locally on one laptop GPU (RTX 4060, 8 GB). A week-1 bake-off on development documents picks the reader. Cloud models serve only as an offline benchmark on synthetic data.

MODEL & TOOL STACK (CANDIDATES; THE BAKE-OFF DECIDES)

ROLE	CHOICE	WHY
Reader (VLM)	**Qwen3.5-4B / 9B** (non-thinking), with Qwen3-VL-8B as the control	Natively multimodal, Apache-2.0, fits 8 GB at 4-bit. No Arabic benchmark yet, so we measure
Second reader	**PP-OCRv5 Arabic** vs Surya 2	Arabic lines with boxes and scores. Every evidence box comes from here
Cloud reference	Gemini 3.5 Flash-Lite / 3.8 Flash	Benchmark ceiling only; synthetic data; never live
Serving	Ollama 0.40 (OpenAI-compatible)	JSON-schema output + token logprobs (checked with images in week 1)
Scoring	scikit-learn logistic regression	Calibrated P(correct) from several evidence signals
Checks	OpenCV · CAMeL Tools · rapidfuzz · Pydantic	Quality gate, Arabic + Kurdish normalization, schemas
App	FastAPI + Jinja2 + HTMX (RTL)	One Python stack, escaped UI, phone camera upload
Data & eval	Playwright · Augraphy · pytest	Render with box labels, degrade, test

Why two readers. Arabic character error rate, print / handwriting: Surya 0.20 / 0.25, PaddleOCR-VL 0.33 / 0.44, Qwen3-VL-8B 0.36 / 0.50. Qwen3-VL boxes were rated poor on 12 of 14 Arabic/French pages (arXiv 2610.25235, Oct 2026 preprint). No reader is reliable alone, so agreement is the evidence.

Week-1 bake-off on ~150 development photos: field accuracy, nulls, JSON validity, VRAM, p95 latency. The decision memo goes in the repo.

SYNTHETIC DATA PIPELINE: FICTIONAL BY CONSTRUCTION

1. **Profiles**
   An LLM generates fictional Iraqi people and shops from our schema: given name, father, grandfather and laqab; mother; Kurdish spellings; governorate, business, numbers. We hand-check 10%, deduplicate and seed every run.

2. **Our own templates, unmistakably fake**
   2–3 HTML layouts each for the ID card, business license and tax card. Each carries a نموذج تجريبي / SPECIMEN watermark, ICAO's fictional issuer "UTO" in the MRZ, invalid number ranges, and no emblems, stamps or security features.

3. **Render with free labels**
   Playwright renders each template, and the DOM gives every field's exact box as ground truth. Handwriting comes from Ruqaa-style fonts plus ~40 fields written by hand.

4. **Plant the hard cases**
   Spelling variants vs real name conflicts, missing name parts, expired or missing documents, the wrong document type, unreadable fields, and instruction text injected into a document.

5. **Photograph badly on purpose**
   Digital: blur, glare, tilt, cut corners, shadows, WhatsApp-style compression.
   Physical: print, laminate, then shoot with 2 phones in 6 conditions (low light, lamp glare, 30–45° angle, finger over a field, cut corner, normal).

Target: ≈260 fictional packets (≈800 document photos), with 40+ physical phone photos in the test set. Generator code, seeds and manifests go in the repo. No real document, ever.

Pre-registered metrics on a sealed test set: safety first, then reviewer time saved

A good result is not "high accuracy". It is **no unsafe auto-pass**, **no invented field**, confidence that means what it says, and fewer fields a person has to touch. Every number below is a target, not a result.

SPLITS (BY FICTIONAL PERSON AND TEMPLATE, SO NOTHING LEAKS)



Development tunes prompts and rules. **Calibration** fits the scorer and threshold. **Validation** gives the MVP's "first results". The **locked test** is sealed as it is created: SHA-256 manifests are committed to git (digital in week 1, phone photos in week 2). It is opened once, at the progress review, and contains ≥40 physical phone photos and one layout the system has never seen.

HOW A FIELD EARNS "ACCEPTED"

Evidence signals
VLM ↔ OCR agreement · crop re-read agreement · value-token logprobs · validators · crop quality

→

Calibrator
logistic regression fitted on the calibration split → P(correct)

→

Threshold
lowest τ whose 95% Clopper–Pearson bound keeps accepted-field error ≤ 2%

Self-reported confidence is weak at separating right from wrong extractions (AUROC 0.54–0.74). Combining perception, layout and validation signals reached 0.90–0.99 (arXiv 2609.20110, preprint). So the model's own confidence is one input, never the gate. Null is a first-class answer: a required field that is null always goes to a human.

COMPARISONS THAT SHOW WHAT EACH PART BUYS

A: classic OCR + regex · B: VLM single pass with its own confidence · **C: Qareena (evidence loop + calibration)** · D: Gemini reference. Results are broken down by clean, digital-degraded, physical, WhatsApp and unseen layout.

SCORECARD: TARGETS FIXED BEFORE WE LOOK AT THE TEST SET

METRIC	WHY IT MATTERS	TARGET
False auto-pass (packets)	A wrong file waved through is the costly error	**0 / 80 → 95% bound ≤ 3.7%**
Error among auto-accepted fields	Can a reviewer trust a green field?	**≤ 2% (95% bound)**
False fills on unreadable fields	"Never invent a field"	**0**
Cross-document conflicts caught	Planted name and number mismatches	**recall ≥ 95% · precision ≥ 85%**
Calibration	"90% means right 90% of the time"	**adaptive ECE ≤ 0.05, plus a reliability diagram**
Bad photo caught at capture	Blur, glare and cut corners fixed before review	**recall ≥ 90%**
Fields a reviewer must touch	The time-saving claim	**reported with CI, no inflated target**
Exact field accuracy (CER too)	Reading quality by field and condition	**reported per slice**
Prompt-injection success	20 documents with planted instructions	**0 accepted**
Latency per packet (RTX 4060)	Usable live, on judges' input	**p95 ≤ 90 s**

Every rate is reported with a Wilson 95% interval and n. After the MVP, a timed study with 3–5 reviewers compares paper checklist vs workbench on seeded packets and measures minutes per file and defects caught.

What could go wrong, and the week-by-week path to a working MVP on 8 November

Each risk has a mitigation and a measurement that tells us whether it worked. The failure cases double as the three-plus documented failure modes for the progress review.

MAIN RISKS

RISK	MITIGATION	HOW WE'LL KNOW
Local model too weak on Arabic handwriting or bad photos	Week-1 bake-off with a model ladder (Qwen3.5-9B → 4B → Qwen3-VL-8B). If a field stays uncertain, route it to a human rather than guess	Accuracy per field & condition
Confidently wrong reads	Multi-signal calibration, crop re-read + second OCR, 5% random audit of auto-passes	False auto-pass, reliability diagram
Prompt injection, hostile files, XSS via extracted text	Model has no tools; schema-only output; instruction-text detector; re-encode with size/pixel caps; escaped UI with CSP	Attack-set success rate
Name variants vs real conflicts confused	Token-level matcher with explicit classes; conservative normalization; ID numbers exact only	Conflict precision / recall
Synthetic-to-real gap; judges' unseen layout	Physical photo tier; reads by label, not coordinates; unseen-layout slice	Results per tier
Live demo fails (GPU, laptop, network)	Fully offline, model pre-warmed, fail-safe is "human review", never "pass"	Rehearsal on fresh input
Model loops or emits broken JSON	Non-thinking mode, schema-constrained decoding, token and time caps. Any failure → human review	JSON-validity & timeout rate
Thin evidence from real users	Reviewer interview in week 1; ask ZainCash for a session; timed study after MVP	Notes & timings in repo

PLAN TO MVP (8+ H/DAY, SOLO)

Week 1 — 23–29 Oct

Foundations & sealed data

Field schema and rulebook v1; profile generator; 3 document templates × 2 layouts; renderer with box labels; degradations. **Seal the digital locked test (hash commit).** Model bake-off → decision memo. Reviewer interview #1.

Week 2 — 30 Oct – 5 Nov

Core, end to end

Intake guard + quality gate + retake messages; VLM reader with schema & logprobs; rules + Arabic name matcher (unit-tested); evidence loop; calibrator & threshold; print & photograph the physical tier and **seal it**; reviewer workbench v0 (Arabic RTL).

MVP — 6 – 8 Nov

First results & submission

Validation-split results with CIs and a reliability diagram; 3 failure modes documented with demo cases; one-page update; rehearsal on fresh input → submit 8 Nov.

After — to review

Toward the progress review

Timed reviewer study, live capture guidance, wider red-team set, then open the locked test once.

REPOSITORY
`github.com/V0ightt/qareena`

Real commit history from today. No keys in git (.env + secret scan in CI). Every model, tool, dataset and AI assistant is listed in **`docs/DISCLOSURE.md`**. AI assistants used: Claude Code (Anthropic), ChatGPT / Codex (OpenAI).